COL774

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f:- True target function $\qquad D = \{x^{(i)}, y^{(i)}\}_{i=1}^{m}$

$Bias(x) = |E_D[f(x) - M(x)]|$ $\quad x^{(i)}, y^{(i)} \sim Dist$

$\hookrightarrow |f(x) - \hat{M}(x)|$

$\hat{M}(x) \hookrightarrow E_D[M(x)]$

$Var(x) = E_D[[M(x) - \hat{M}(x)]^2]$

Net (squared) error = $Bias^2$ + Variance + Noise

Random Forests:- collection of decision trees.

$M_1 \ldots M_K$

$\hookrightarrow M_{avg}(x) = \frac{1}{K}\sum_{k=1}^{K} M_k(x)$

Data $= \{x^{(i)}, y^{(i)}\}_{i=1}^{m}$

$D_1 \quad D_2 \quad \ldots \quad D_K$

$\forall k \in \{1 \ldots K\}$

$D_k = \{\}$

for $i = 1$ to m {

$D_k = D_k \cup \{x^{(i)}, y^{(i)}\}$

$\hookrightarrow (x^{(i)}, y^{(i)}) \sim$ Unif Data

}

1 ... 100

1, 3, 5, 16, ... 3, ... 99

$\forall k \in \{1 \ldots K\}$

Learn M_k on D_k

$M_{avg}(x) = \frac{1}{K}\sum_{k=1}^{K} M_k(x)$

1. \# of trees
2. $\{x_1, \ldots x_n\}$ $\hookrightarrow$ Randomly choose r sized subset $(r < n)$ S of variables to split on
   $\hookrightarrow$ chose best from S (at every decision node)
3. $E_k = \frac{1}{[|x^{(i)}, y^{(i)}| : (x^{(i)}, y^{(i)}) \notin D_k]} \sum_i \mathbb{1}\{M_k(x^{(i)}) \neq y^{(i)}\} \cdot \mathbb{1}\{(x^{(i)}, y^{(i)}) \notin D_k\}$